

SARTHAK MAKKAR

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SUMMARY

B.Tech Information Technology student (graduating July 2027) with production backend experience and two independently deployed full-stack systems. Strong in API design, relational data modeling, and system reliability patterns (caching, rate limiting, authentication, RBAC), with hands-on Agile/Scrum delivery experience. Seeking Software Development Engineer / Backend / Full Stack roles.

SKILLS

Backend: FastAPI, Node.js, Express.js, TypeScript, REST APIs, PostgreSQL, Redis, SQLAlchemy, BullMQ, JWT Auth, OAuth 2.0, Docker

Frontend: React, Next.js, JavaScript, Tailwind CSS, Framer Motion

Core CS: Python, C++, SQL, Data Structures & Algorithms, Agile/Scrum, Git, Linux

Applied AI (from internship): RAG Pipelines, LLM Integration & Fallback Design, Vector Databases (Qdrant, FAISS)

EXPERIENCE

Technology Intern — ShortHills AI

Jun 2026 – Aug 2026 | Gurugram, India

- Developed a production RAG pipeline's vector store from FAISS to Qdrant and tuned chunking strategy within an Agile/Scrum sprint workflow, raising retrieval completeness to 92%.
- Built multi-provider LLM fallback handling (Gemini, OpenRouter, GPT-OSS-120B) for graceful degradation under API rate limits, sustaining zero hard crashes under peak request volume.
- Led a web-scraping and report automation platform achieving 91.08% extraction accuracy across 5 enterprise client sites.

PROJECTS

ReachInbox — Email Job Scheduler — reach-inbox-topaz.vercel.app

- Engineered a production-grade email scheduling backend (Node.js, TypeScript, Express, BullMQ, Redis) using persistent delayed-job scheduling in place of cron, surviving process restarts via a boot-time reconciliation service with zero duplicate or lost sends.
- Implemented per-sender rate limiting with atomic Redis counters and automatic job rescheduling on limit breach, paired with live Slack OAuth 2.0 alerts deduplicated via distributed locking to prevent notification spam.
- Designed idempotent job processing using atomic conditional database updates, guaranteeing exactly-once email delivery under concurrent worker execution.
- Built a Next.js/Tailwind dashboard with Google OAuth authentication, CSV-based bulk recipient upload, and real-time scheduled/sent tracking; deployed full stack across Render, Vercel, Neon PostgreSQL, and Redis.

Earnease — Earned Wage Access Platform — earnease-seven.vercel.app

- Built a two-tier eligibility rules engine (non-negotiable hard-fail rules versus negotiable rules routed to HR) as a standalone, independently testable module decoupled from HTTP handlers.
- Modeled a 6-state request lifecycle (Pending, Auto-Approved, Auto-Rejected, Escalated to HR, HR-Approved, HR-Rejected) with append-only audit logging kept separate from mutable request state.
- Implemented role-based access control (Employee/HR) with per-request database-backed verification, and a wage-accrual engine preventing duplicate advances within a single pay cycle.
- Deployed full-stack (FastAPI, PostgreSQL, React, Framer Motion) with calendar-day rate limiting and bcrypt password hashing hardened against long passphrase inputs.

ADDITIONAL PROJECTS

- Quantum Scraper — Fault-tolerant async extraction pipeline (Python, AsyncIO, FastAPI) combining semantic DOM analysis, JSON-LD parsing, and LLM-assisted extraction; achieved 91.08% average accuracy across 5 enterprise manufacturer sites — quantum-scraper-beryl.vercel.app
- Ask-Wiki — RAG-based Wikipedia chatbot evaluated across 7,500 test queries; 89% weighted composite score (RAGAS/DeepEval), 92% precision at 500-query threshold — huggingface.co/spaces/sarthak1001/ai-wikipedia-rag

ACHIEVEMENTS

- Hack hazards 2024 — Placed in the Top 15 teams out of the full competing field.

EDUCATION

Bhagwan Parshuram Institute of Technology (BPIT) — Delhi, India

B.Tech, Information Technology — CGPA: 8.495 | 2023 – 2027 (Graduating July 2027)

CBSE Class 12: 88% | Class 10: 93.2%